

MATTER 4-CHANNEL SMART CONTROLLER

Office Automation Device Specification & BOM

1. Device Overview & Core Features

- Connectivity: Native Matter over Wi-Fi standard. Connects seamlessly with Home Assistant, Apple Home, Google Home, and Alexa entirely via local network control.
- Control & Isolation: 4 independent relay channels for appliance switching, integrated with optocouplers (DIP-4) to insulate logic circuits from high-voltage noise.
- Power Options: Flexible power grid accommodating integrated AC-DC (Hi-Link module).
- Enclosure: 3D-printed robust bottom base frame with a transparent CNC-cut acrylic top lid for effortless visual diagnosis.
- Has options to integrate Motion sensor and Biometric fingerprint sensor. We are planning to give fingerprint sensor for server room and normal motion sensor for other room.

2. Hardware Layout & Reference Renders

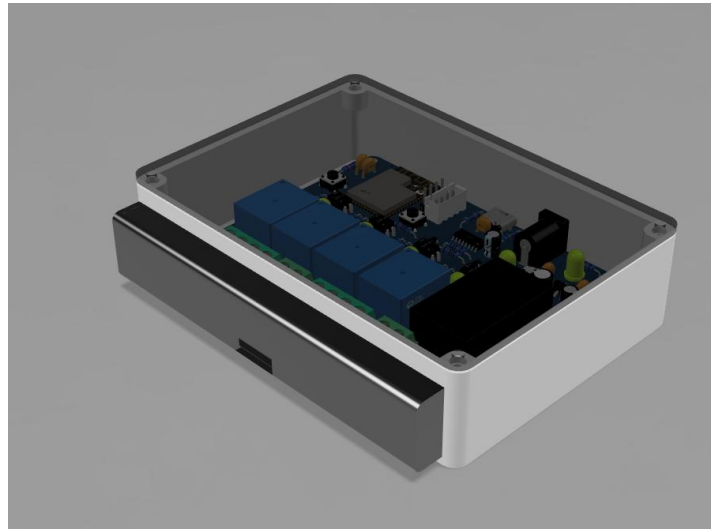


Figure 1: 3D Rendered Isometric Enclosure View

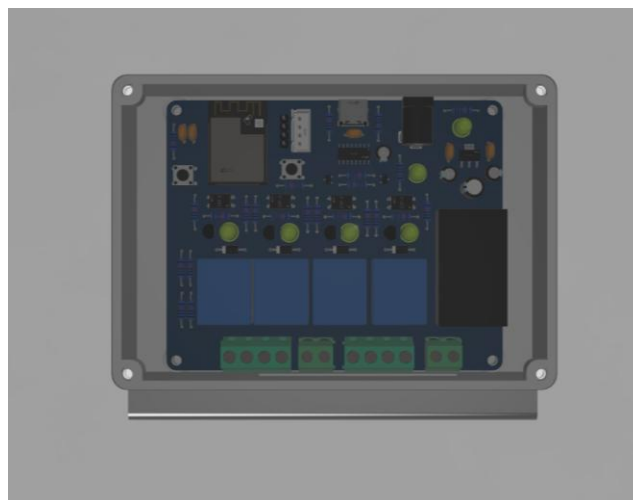


Figure 2: 3D Rendered Top Internal View

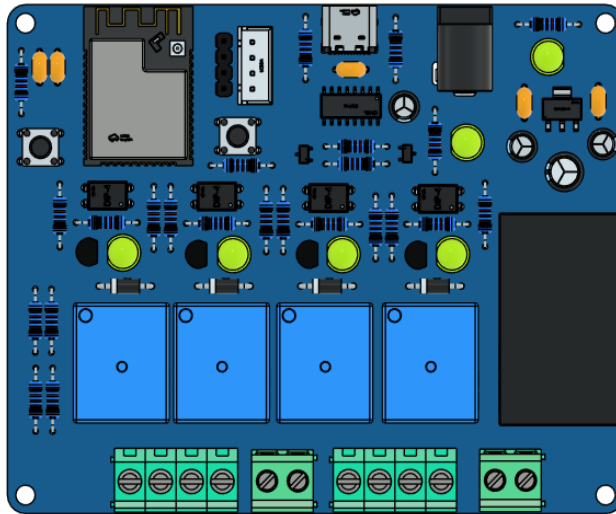


Figure 3: Custom 4-Channel PCB Layout Component Placement

3. Production Bill of Materials (BOM)

Item Description	Qty	Unit Spec	Total / Source
PCB Order Fabrication	5	—	600 BDT (JLPCB)
PCB Component Set (ESP32, Relays, Hi-Link, ICs, Optocouplers)	5 set	1,200 BDT	6,000 BDT (Local/Sourced)
3D Printed Enclosure Base	5 set	50g	250g Total (In-House)
Acrylic Top Sheet Cover	2 set	200 BDT	400 BDT (Daraz)